James Anderson

PERSONAL PROFILE

I am an electronic engineer with specific interests in ML, automation, embedded systems, audio tech, and digital design technology. I have experience working within large and small multidisciplinary teams. I am enthusiastic about the opportunity to better myself by developing technical ability within different industrial settings with an ethos conducive towards a forward thinking, innovative, development environment. I am passionate about design in all its forms as well as utilising it to provide significant, impactful benefits.

RELEVANT PROJECTS

Abelon

Track Cycling Telemetry Device

With the ambassador for this project being Sir Chris Hoy, this was an exciting first project to be placed on at Abelon. I furthered my experience with Espressif's ESP-IDF on an ESP32. The product I was developing was a track cycling telemetry device which communicated with BLE sensors (heart rate, cadence, power) via NimBLE and logged the data to flash. Screens are not allowed on bikes, so the device would relay data via a bluetooth earpiece.

Aker Subsea Electronics Module

By far the largest project I have worked on to date (1,000,000+ lines of C, C++), an Aker Subsea Electronics Module. Requiring me to port legacy firmware to new hardware with the need to maintain behaviours for backwards compatibility. Oil and gas subsea hardware goes into the field for a very long time due to the remote, extreme environment, due to this nature, the technology implemented is over 20 years old, finding comments from as early as 1997! I've developed my skills to learn Blackberry's QNX (an embedded Unix microkernel based os), more complex architecture, strict testing procedures adhering with Aker's standards, version control and working within a larger team.

Greengage

I built and developed an IoT sensor enabled data platform for livestock farming to measure environmental impact, energy efficiency, disease sniffing and welfare. Firmware for sensors that measured NH_3, CO_2, CH_2O, CH_4, temperature, lux, humidity, movement, CCTV camera vision analysis, FFT frequency analysis - this was developed in partnership with Newcastle University.

This includes a diagnostics system I designed. This involved reporting to a new MQTT topic *diagnostics* to relay essential operational data. This allowed the devices to communicate performance metrics and if maintenance was needed e.g. battery health, probe/camera status. AWS data pipelines were built using MQTT, RDS, S3 and Lambda. This connected to a web app which was developed by two contractors.

I also developed a device using an ESP32 to communicate via serial comms to a 3rd party device and parse the response into JSON to send data as well as object detection algorithms for "chicken cluster detection" using pytorch. I also have learned a lot about working in a small business in a tight knit group and designing with a budget in mind. Being creative and utilising what I currently have. I've also had the opportunity to work closely with other departments such as RMA's in operations and with logistics lending a hand where I can. Providing technical sales support and assisting installations on site.

Master's Project	Robotics, pick and place automation utilising the Robotnik Summit XL Steel, UR5 arm, and Robotiq's 3-finger gripper. Simulation tested with ROS, RViz, and Gazebo. My focus was on the operation of the mecanum wheels and the navigational stack of the robot.
Formula Student	Formula Student is an engineering competition where different university teams compete in building a car, for us an EV where I worked on the electric drivetrain.
Advanced Reading Project WiFi Based Monitoring of Activities of Daily Living	My advanced reading project involved creating a systematic literature report and to utilise an ESP32 microcontroller to collect channel-state-information (CSI). This data is then analysed and can be used to track human movements.

EDUCATION

MEng

Electrical and Electronic Engineering

Heriot-Watt University

5th Year: MEng Awarded
4th Year: BEng (hons) 1st Awarded
3rd Year (1st): Core Modules
2nd Year (2:1): Secondary Modules.
1st Year (1st): Introductory Modules.

WORK EXPERIENCE

Software Engineer (Abelon)
Electronic and AI Engineer (Greengage)
Innovation Ambassador (Business Enterprise Heriot-Watt University)
Kitchen Staff & Front of House (Hau Han)
Deliveroo (Cycling), Edinburgh
Crew Member Mcdonald's, Cameron Toll, Edinburgh
Paper Round (Distributor of Stockton & Billingham, Herald & Post)

SKILLS/HOBBIES

Sound Engineering / Production, Photography, Guitar, Bass, Drums, Skateboarding, Skiing.
Bronze Award DofE, Little Princess Trust (1ft hair donation + £203)
Ben Nevis Climb for Macmillan Cancer Support (Collectively raised over £11k)

REFEREES

Available on request.